AC9M6A01 • Year 6 Mathematics

Growing Patterns and Rational-number Rules

Connect visual growth, term values and direct or recursive rules

We are learning to…

Students identify what changes and what remains constant, represent patterns in tables and diagrams, use recursive and direct rules and test whether a rule generates every term.

Represent | Reason | Calculate | Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

recognise and use rules that generate visually growing patterns and number patterns involving rational numbers

Connect a growing tile pattern to a rule

term n	**1**	**2**	**3**	**4**	**n**
tiles	**5**	8	11	14	3n + 2
growth		+3	+3	+3	constant +3

The recursive rule adds 3 each term. The direct rule 3n + 2 gives any term without generating all previous terms.

Extend patterns involving fractions and decimals

0.5, 1.0, 1.5, 2.0

add 0.5; term = 0.5n

3/4, 1, 1 1/4, 1 1/2

add 1/4; term = n/4 + 1/2

visual border growth

count repeated section plus fixed corners

digital table

test many terms after defining rule

A rule must fit the visual structure and all known terms. Equivalent forms of a rational rule may look different but produce the same values.

Build and annotate the model

Represent growing patterns and rational-number rules and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Rule describes only the next term:** A direct rule should work for any term number.
- **Constant part omitted:** Visual patterns often contain repeated growth plus fixed pieces.
- **Rule accepted from two terms only:** Test several terms and the visual structure.
- **Term number confused with term value:** Keep input n separate from output.

Quick check

- Find the next two terms.
- Write a recursive rule.
- Test 3n + 2.
- Describe fixed and growing parts.
- Extend a decimal pattern.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.

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